

The model tester: what counts as success

Read this page first on every update. PASS refers only to the named check and experiment; unresolved checks cannot become green because a solver converged.

Check	Success means	Current result
Pension accounts	Balance at every solved date under the pinned fiscal gate.	PASS; max scaled residual 3.34e-10
Housing equilibrium	Demand equals supply under the pinned market gate.	PASS; max relative residual 3.56e-05
Reproducibility	Saved mapping replays and checkpoint reloads correctly.	PASS
Calibration fit	Every target and parameter visible; economically material misses resolved or explicitly assessed.	NEEDS WORK; complete tables follow. A small total loss alone is insufficient.
2023 utility comparison	Old and new each recalibrated to the same 2023 targets, fiscal closure and numerical gates.	MISSING; current stationary tests freeze parameters and final preference.
Policy functions	Feasible choices and probabilities; occupied reversals explained and sensitive states checked under numerical refinement.	OPEN; plotting objects clarified. Global policy accuracy is not certified.
Historical transition	Correctly timed, comparable data overlays; shock estimated; observed and imposed moments distinguished.	NEEDS WORK; birth-count shape misses. Household path is conditioned on data.
Terminal horizon	Approved tail checks pass and the dates of interest remain stable when the horizon is extended.	NOT PASSED
Policy readiness	All above resolved, with approved population, entry, fiscal and geographic closure.	NOT READY

Which models are being assessed?

Maintained sequential fertility architecture. Old utility has a per-child housing floor; new utility has an equivalence scale and a first-dependent housing floor. Both retain linear child reward. All cases below balance pensions.

Experiment	What was varied / targeted	Interpretation
Early stationary candidate	New utility; early-period working targets; fertility normalized to 2.1.	Loss 1180.225. Candidate, not certified SMM.
Static old / new, late preference	Same structural parameters, final preference and housing supply; balanced pensions.	Losses 1335.720 / 1361.330. Neither recalibrated.
Six-date historical diagnostic	New utility; illustrative announced preference path, 2007-2027.	Converged finite path. Neither fitted history nor horizon-certified policy benchmark.

Early stationary candidate: every target

Working objective = 1180.225; 12 scored restrictions plus separate fertility normalization. Nine structural coordinates; the preference intercept normalizes fertility. Shaded losses exceed 4 (gap exceeds 2 objective scales): a review screen, not a confidence test; some scales are synthetic. Not certified SMM.

Moment	Target	Model	Gap	Weight	Loss
Initial model completed fertility	2.1	2.1	2.5234e-05	--	--
Childless women, ages 40-44	0.19828	0.19079	-0.0074894	35532	1.993
Exactly one child among mothers, ages 40-44	0.21366	0.22038	0.006729	26953	1.2204
Period mean first-birth age	25.976	26.314	0.33767	139.83	15.944
First births at age 30+	0.24928	0.24821	-0.0010698	13866	0.015868
Wealth / annual gross labor earnings	6.1459	5.9809	-0.16494	7.5951	0.20663
Annual bequests / aggregate wealth	0.0088	0.011453	0.0026526	5.1653e+06	36.343
Old wealth/income p90 / median, ages 76-84	3.5159	3.9569	0.44099	10.616	2.0646
Mean occupied rooms, capped at 9	5.5611	5.9204	0.3593	128.02	16.527
Ownership, heads 30-55	0.64833	0.60481	-0.043527	2339.4	4.4322
First-birth room response, -1 to +3	0.72025	0.52856	-0.19169	137.57	5.0546
Rooms: 3+ versus 1-2 resident children (model dependent proxy)	0.34707	0.1864	-0.16067	280.53	7.2415
Recent-parent ownership gap	0.1629	-0.037746	-0.20064	27056	1089.2

Static old / new at the final preference: every target

Diagnostic losses = 1335.720 / 1361.330. Fertility near 1.40 is the stationary outcome at the retained final preference, not a recalibrated 2023 fit. Shading flags gaps exceeding 2 objective scales, not statistical rejection. Targets differ from the early system; do not compare these losses to its loss.

Moment	Target	Old	New	Old gap	New gap	Weight	Old loss	New loss
Completed-fertility measure	1.918	1.4	1.4092	-0.51801	-0.50875	1425.7	382.58	369.03
Childlessness	0.188	0.33434	0.33219	0.14634	0.14419	17181	367.93	357.21
Mean age at first birth	26.045	28.634	28.606	2.589	2.5618	44.444	297.91	291.68
First births at 30+	0.26033	0.38364	0.38211	0.12331	0.12178	10000	152.06	148.31
First-birth room response	0.72025	0.3617	0.35744	-0.35854	-0.3628	137.57	17.684	18.107
Rooms: 3+ vs 1-2 children	0.3677	0.38386	0.23114	0.01616	-0.13656	2958.5	0.77256	55.168
Parent ownership gap (old definition)	0.16766	0.11335	0.1103	-0.054308	-0.057358	14230	41.968	46.815
Ownership (old definition)	0.57547	0.5884	0.58892	0.012925	0.01345	1207.8	0.20178	0.2185
Mean occupied rooms	5.78	6.448	6.4438	0.66802	0.66378	11.973	5.343	5.2755
Wealth / annual labor earnings	6.8731	6.2095	6.2092	-0.66356	-0.66387	6.2877	2.7685	2.7711
Annual bequests / wealth	0.0088	0.012327	0.012332	0.0035269	0.0035315	5.1653e+06	64.252	64.419
Old wealth/income: p90 / median	3.4481	3.6463	3.6504	0.19819	0.20226	56.96	2.2374	2.3301

Parameters: estimates, restrictions and bounds

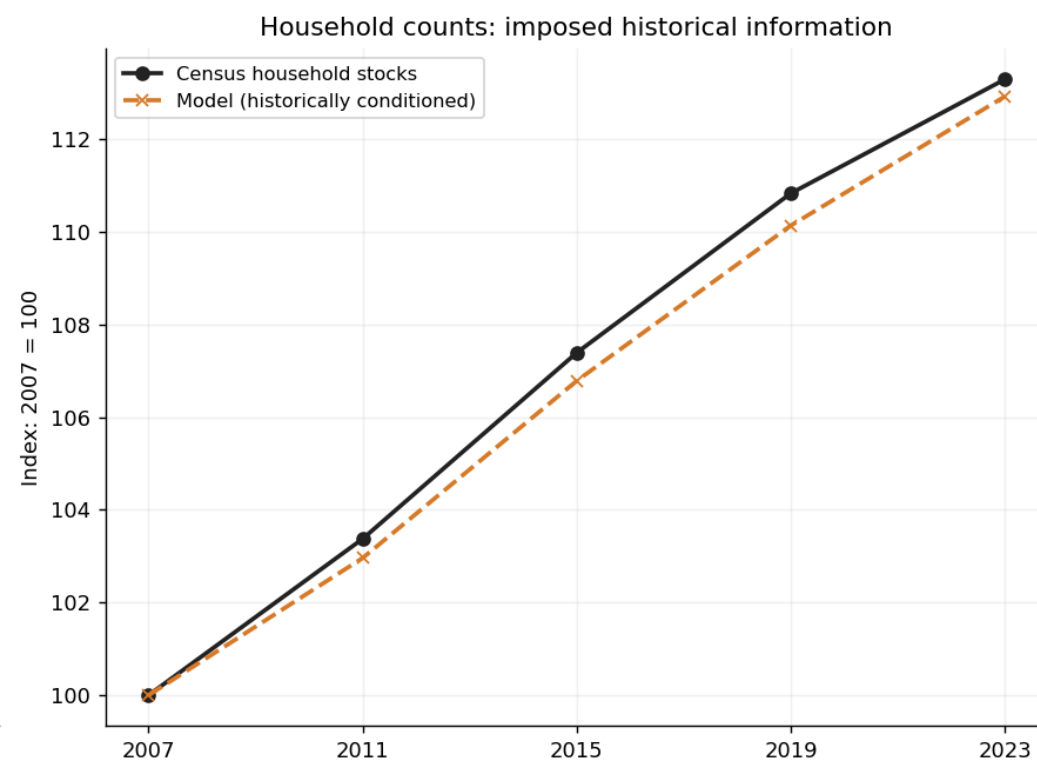
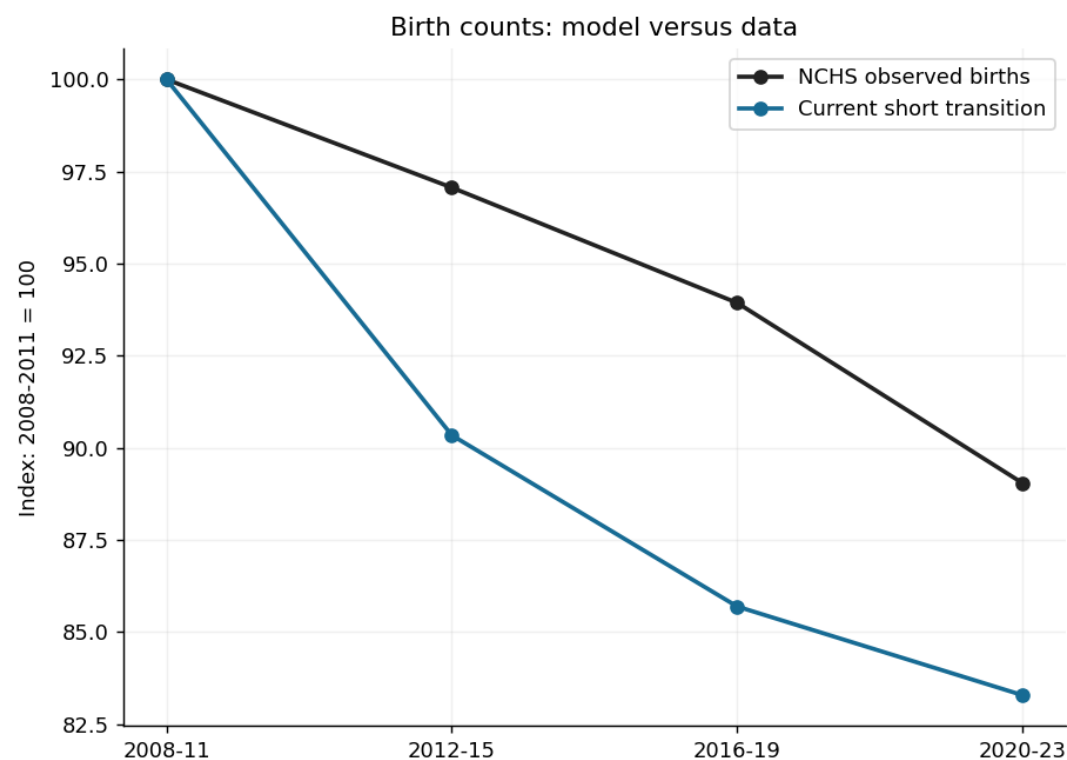
Static columns are inherited, fixed diagnostic inputs. Early estimates are provisional. Bound proximity follows the saved search flags. Pension amount is an endogenous budget outcome; annual beta is raised to the fourth power in four-year periods.

Parameter	Static old	Static new	Early candidate	Range / restriction	Near bound?
beta_annual	0.99528	0.99528	0.99545	[0.94, 0.9995]	No / fixed
kappa_fert	2.1682	2.1682	1.8057	[0.02, 50]	No / fixed
kappa_fert_continuation	1.7708	1.7708	1.6254	[0.02, 50]	No / fixed
chi	1.0537	1.0537	1.075	[0.1, 5]	No / fixed
H0	8.4348	8.4348	7.4778	[0.2, 80]	No / fixed
theta0	0.57035	0.57035	0.44419	[0, 8]	No / fixed
theta1	0.10372	0.10372	0.11955	[0.02, 16]	Yes: theta1, all three
first_birth_fixed_cost	4.6197	4.6197	4.4745	[0, 8]	No / fixed
h_P	0.71685	0.71685	0.92045	New [0.10,2.30]; old = jump + slope	No / fixed
hbar_child_rooms	0.24708	0	0	Old [0.10,1.80]; new fixed at zero	No / fixed
psi_child	-0.032	-0.032	0.29349	Fixed in static tests; early normalized to 2.1	No / fixed
payroll_tax	0.179	0.179	0.179	externally fixed	No / fixed
pension_period	2.0464	2.0464	2.0464	Derived from payroll budget	No / fixed
housing_supply_elasticity	0.63	0.63	0.63	externally fixed	No / fixed
tenure_choice_kappa	0.005	0.005	0.005	externally fixed	No / fixed
alpha_cons	0.733	0.733	0.733	externally fixed	No / fixed
sigma	2	2	2	externally fixed	No / fixed
Old first-child jump	0.46977	--	--	Old [0,0.5]; new uses h_P	No

Transition versus observed data

Birth-count shapes are compared on aligned four-year blocks; household counts are historically conditioned.

What the short transition matches - and what it does not

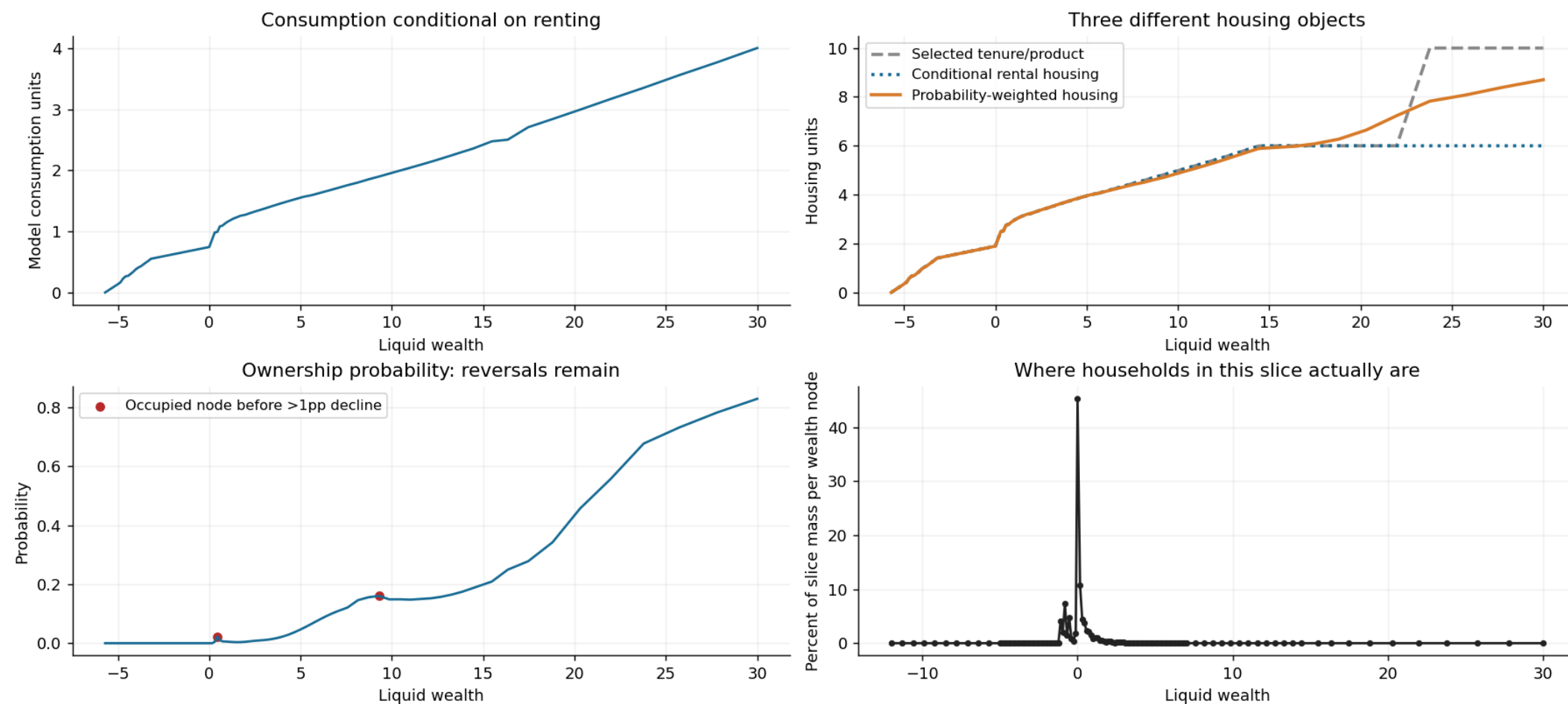


Birth blocks follow the established decision-year + 1 to + 4 clock. Each birth series is normalized to its first block. This compares shapes, not levels or female TFR; national data versus model geography remains an approximation. Household counts enter the historical conditioning and are not independent validation. Sources: NCHS final reports; Census HH-3.

Policy objects and household occupancy

Current 2023 checkpoint: read-only extraction, zero new Bellman solves; numerical reversals remain under investigation.

2023 policy diagnosis: age 30, income state 5 ($z=0.328$)



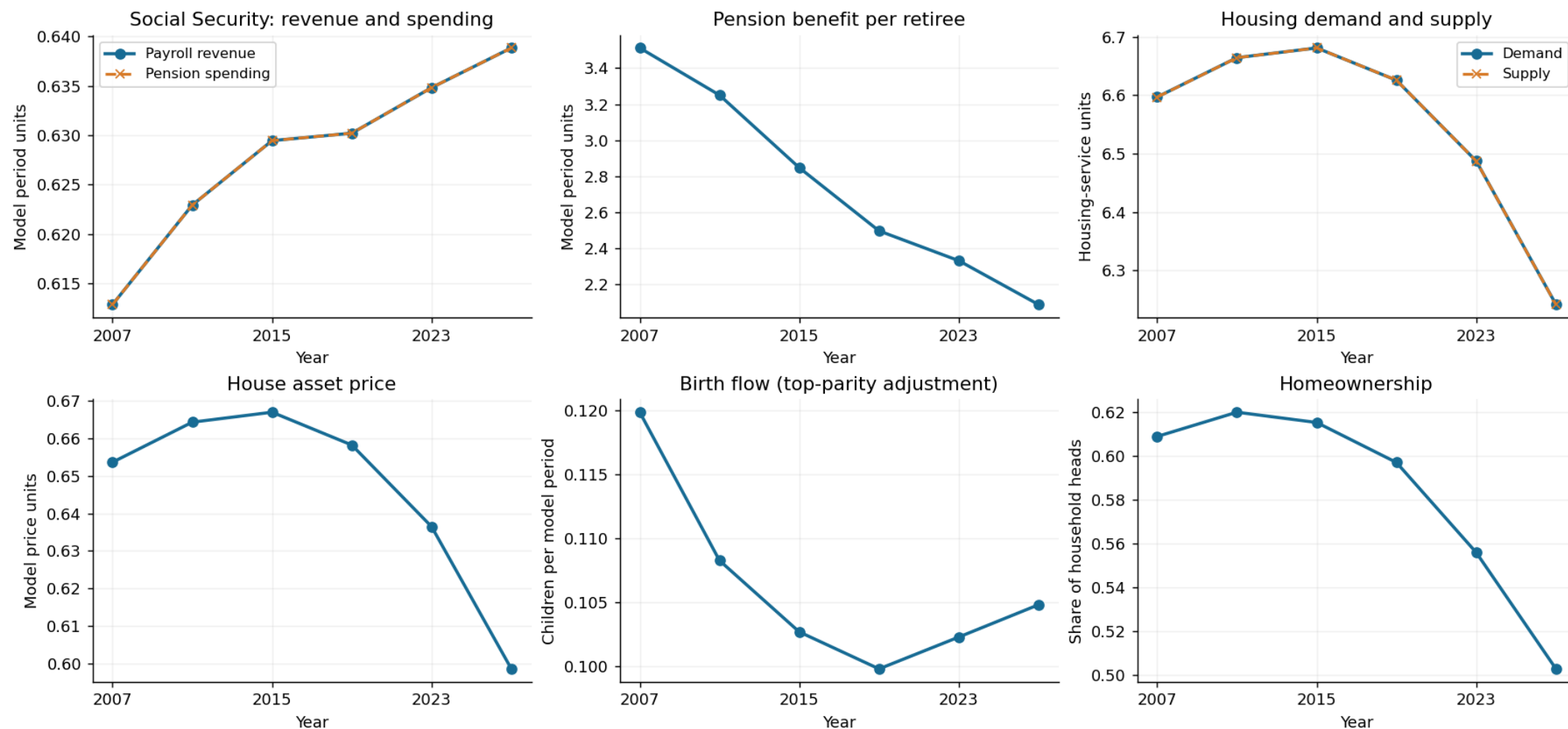
Slice chosen by the largest mass-weighted ownership decline across the inspected age-30/42 income slices.

No smoothing or new solve. Probability-weighted housing corrects the object being viewed; it does not certify the underlying numerical solution.

Transition and pension accounts

Six dates, 2007-2027; new utility; illustrative shock; terminal horizon remains unchecked.

New utility and balanced pensions: solved six-date transition



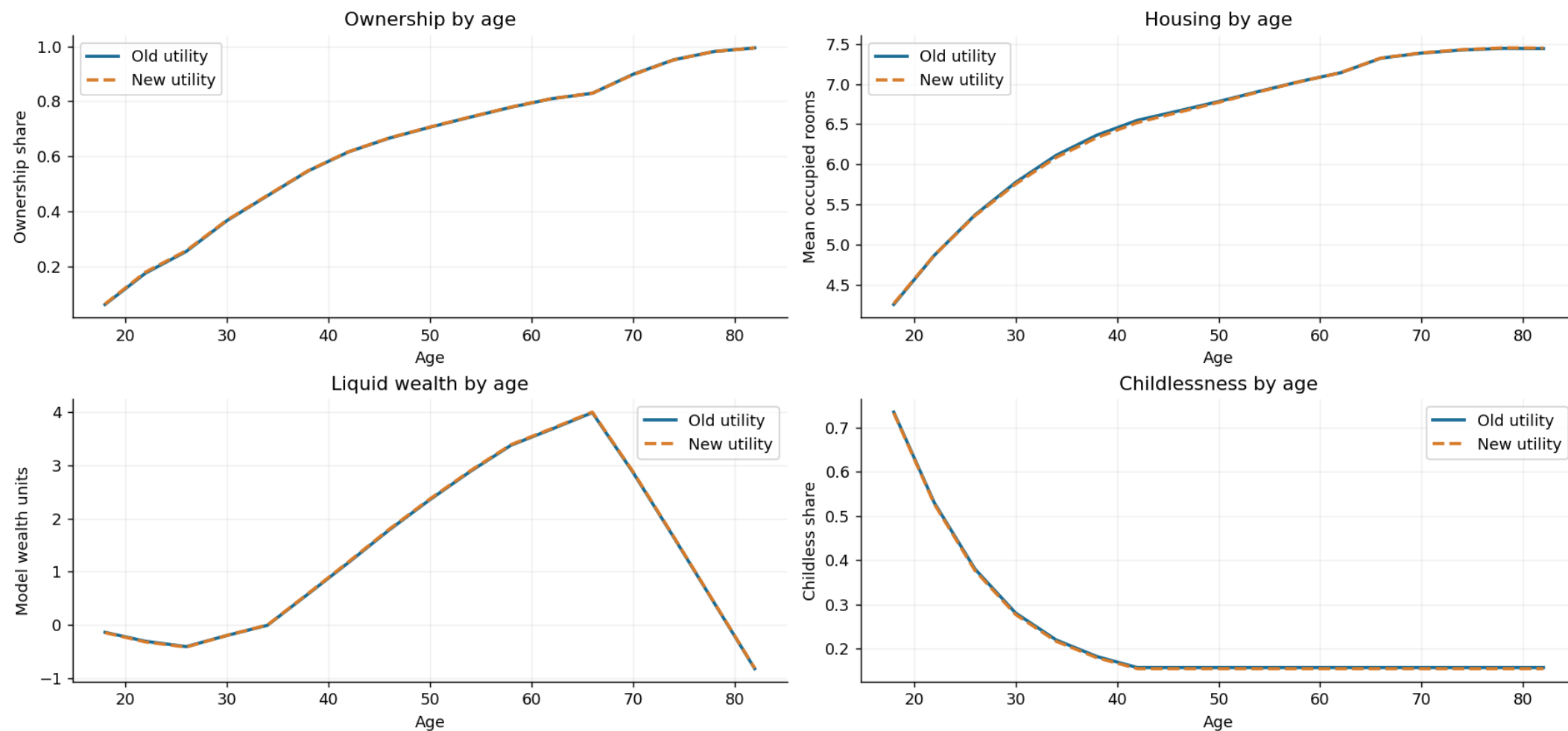
Illustrative announced preference shock; no fitted historical path or horizon certification.

These are model outcomes, not policy effects or population forecasts. Benefits and quantities use four-year model periods.

Direct effect of the utility change

Pre-2007 stationary diagnostics; same parameters and preference; balanced pensions in both cases.

Direct utility comparison: both pension budgets balanced

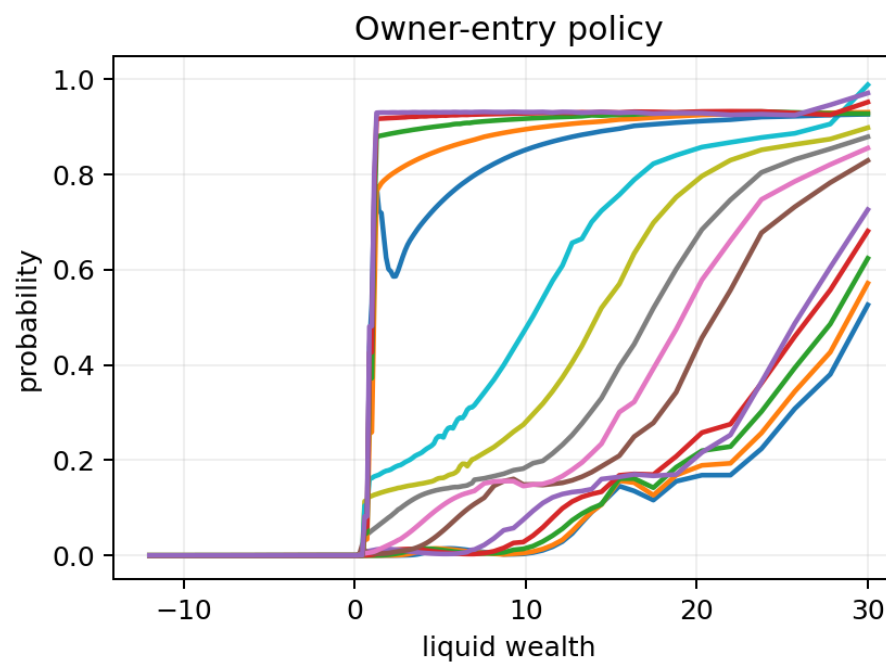
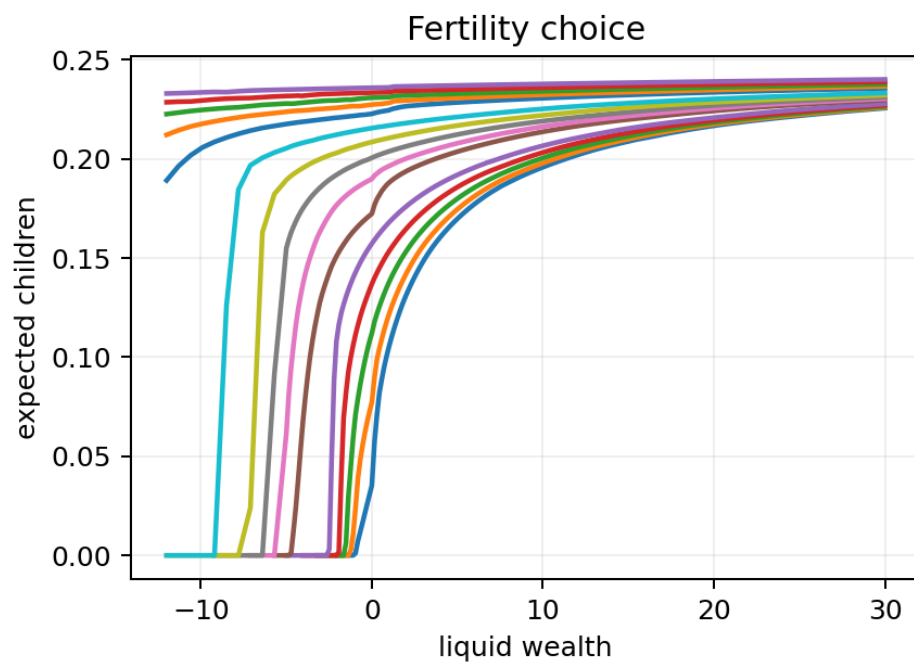
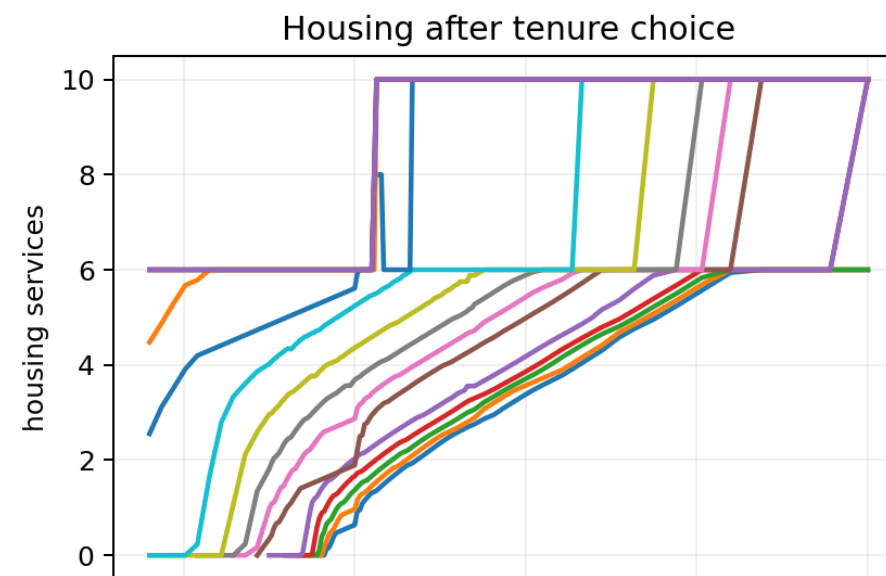
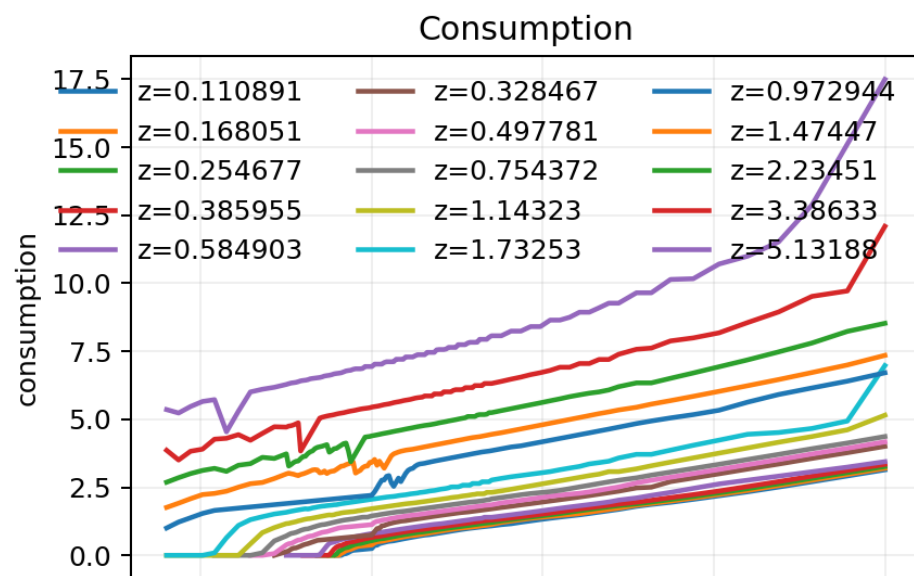


Same structural parameters and fixed fertility preference; separate stationary housing equilibria.
Pre-2007 diagnostic. A matched 2023 recalibration is a separate experiment.

2023 policies: childless renter, age 30

2023 slice of the six-date transition; original saved diagnostic, unchanged.

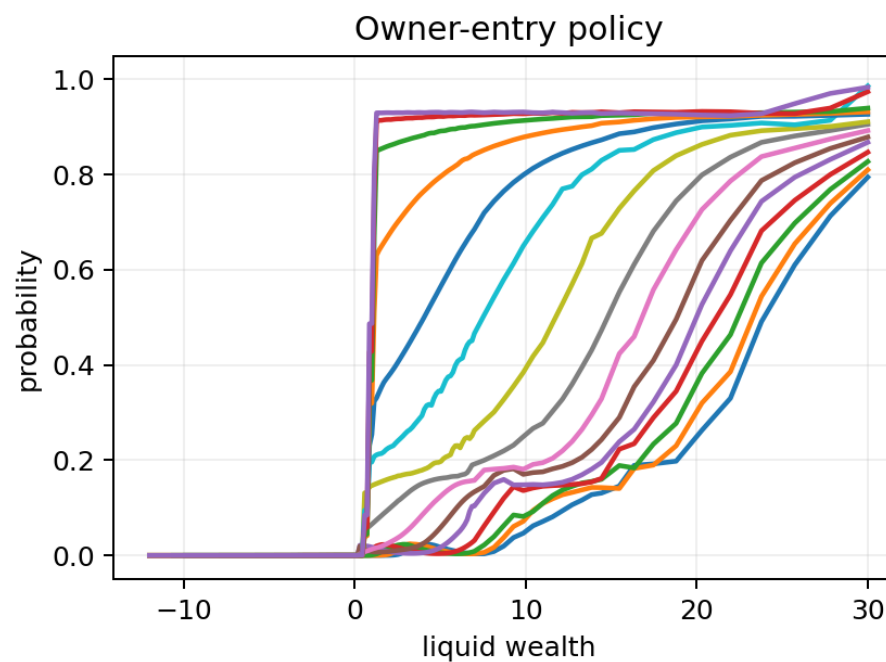
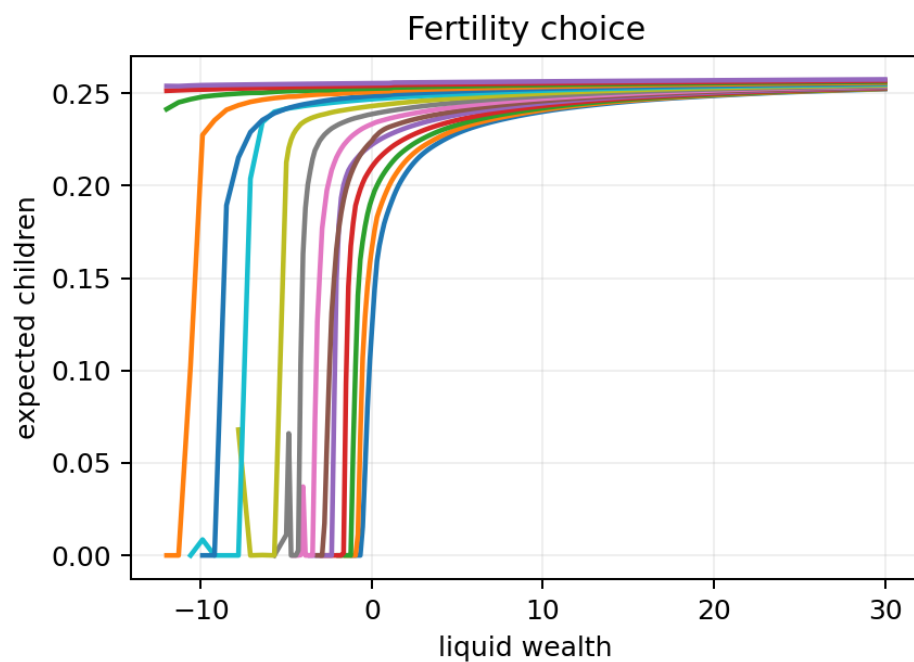
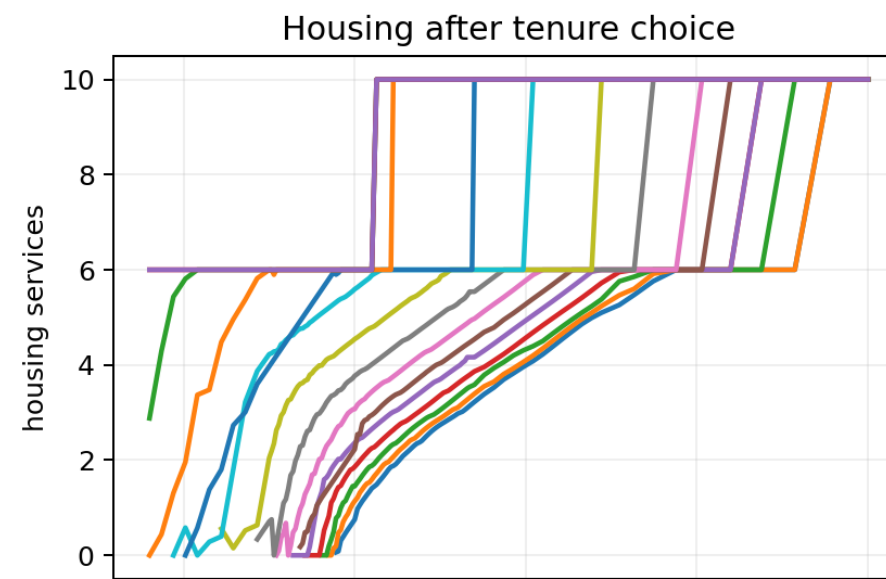
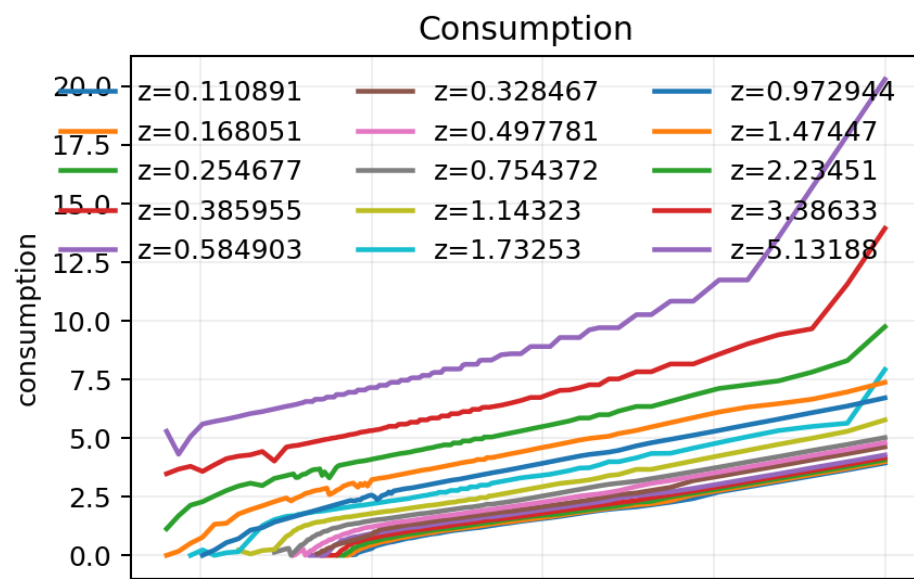
Childless renter policies at age 30



2023 policies: childless renter, age 42

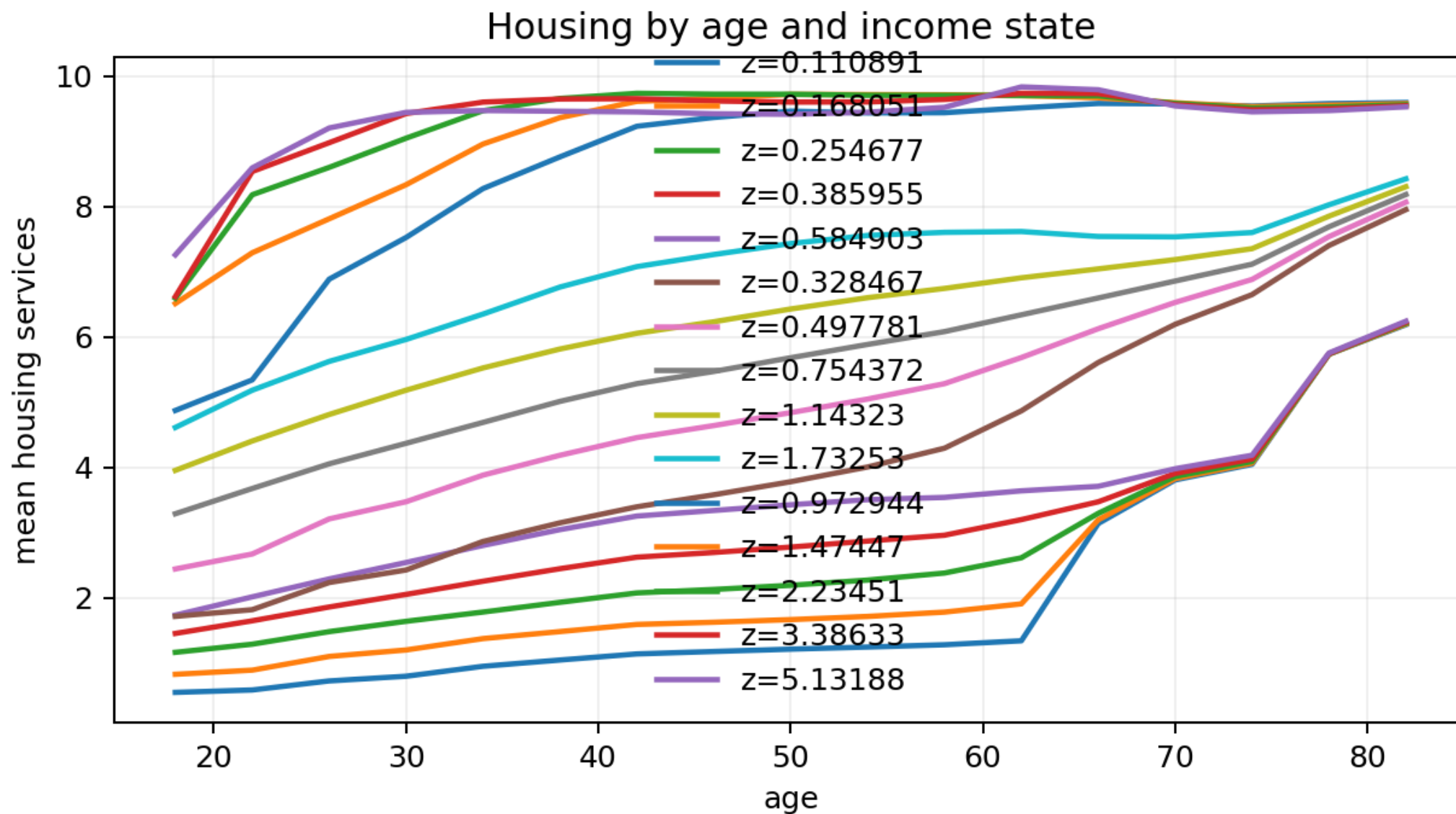
2023 slice of the six-date transition; original saved diagnostic, unchanged.

Childless renter policies at age 42



Housing across ages and income states

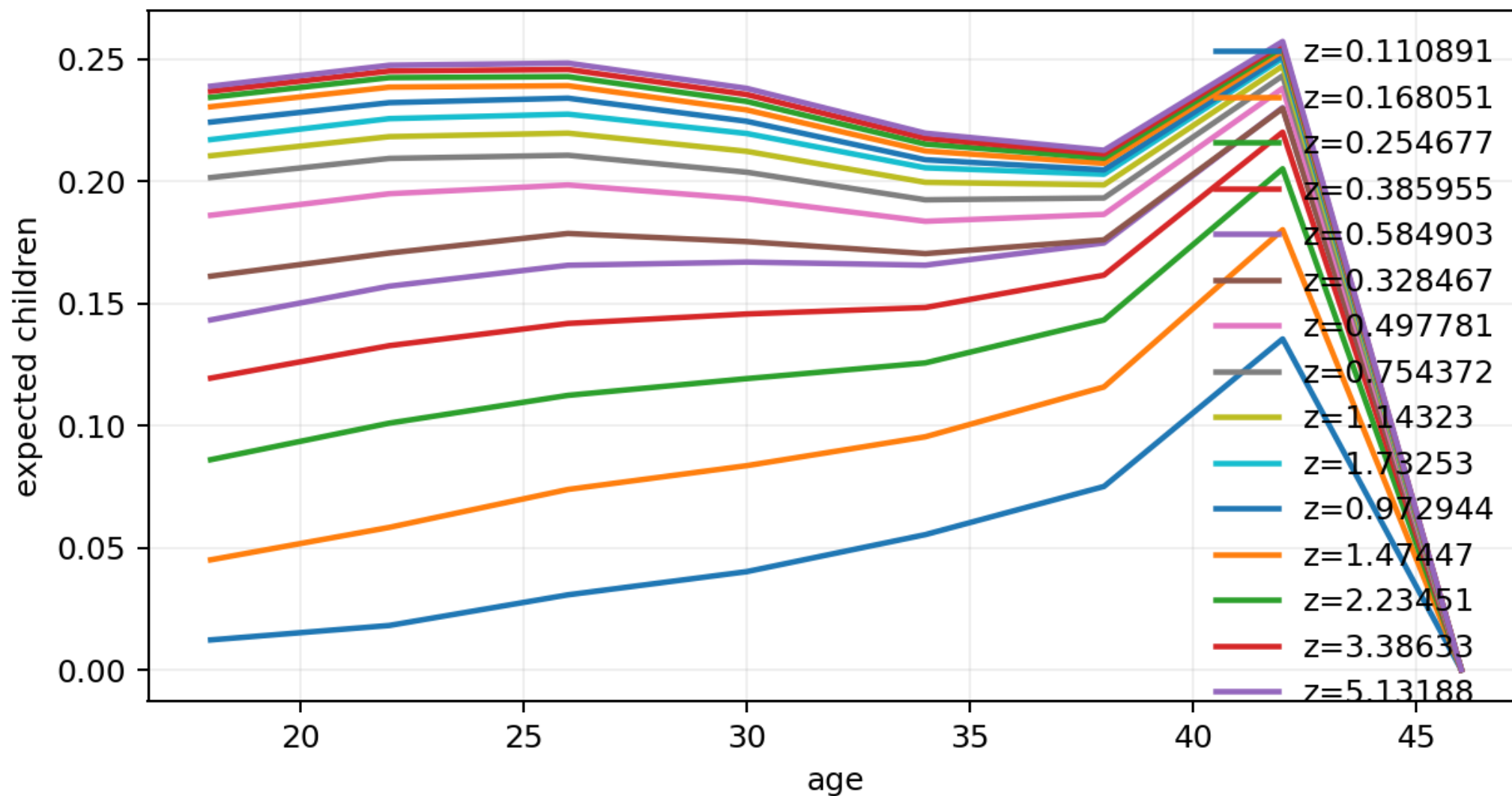
2023 slice of the six-date transition; original saved diagnostic, unchanged.



Fertility across ages and income states

2023 slice of the six-date transition; original saved diagnostic, unchanged.

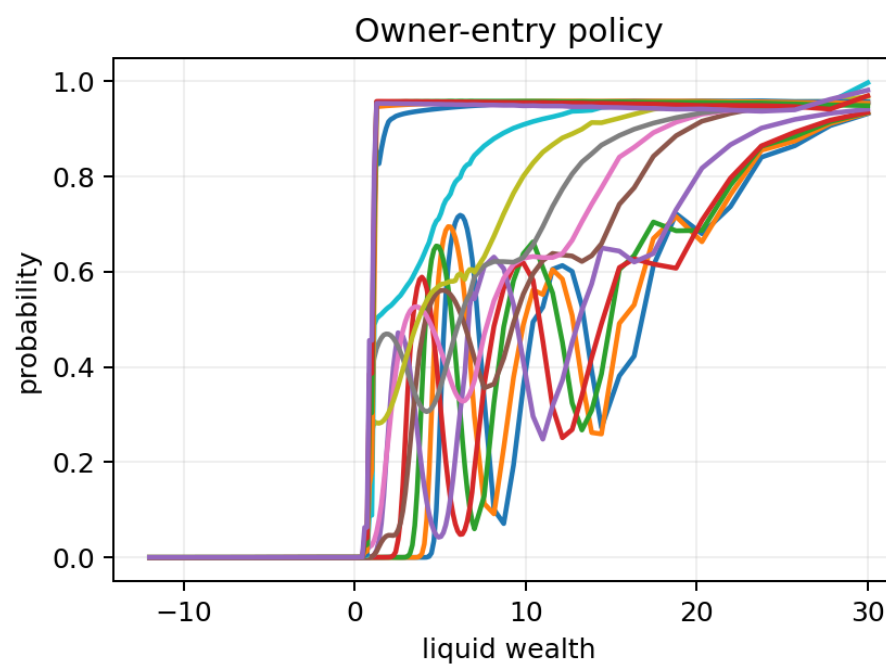
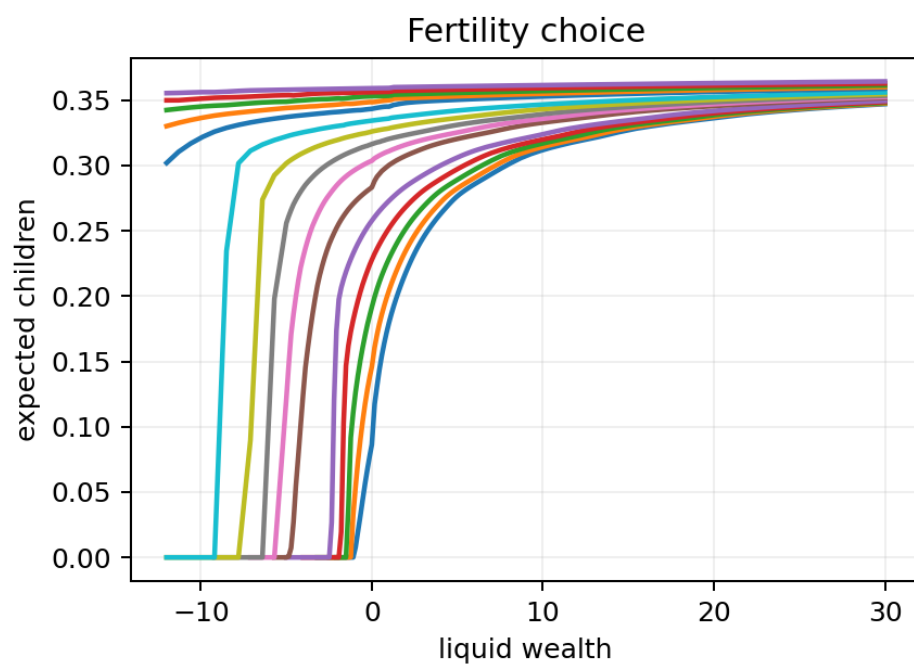
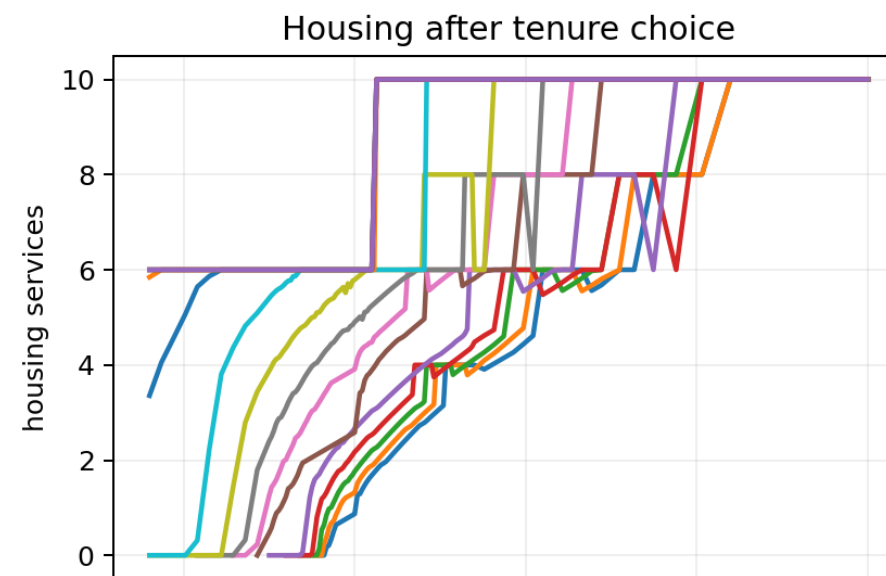
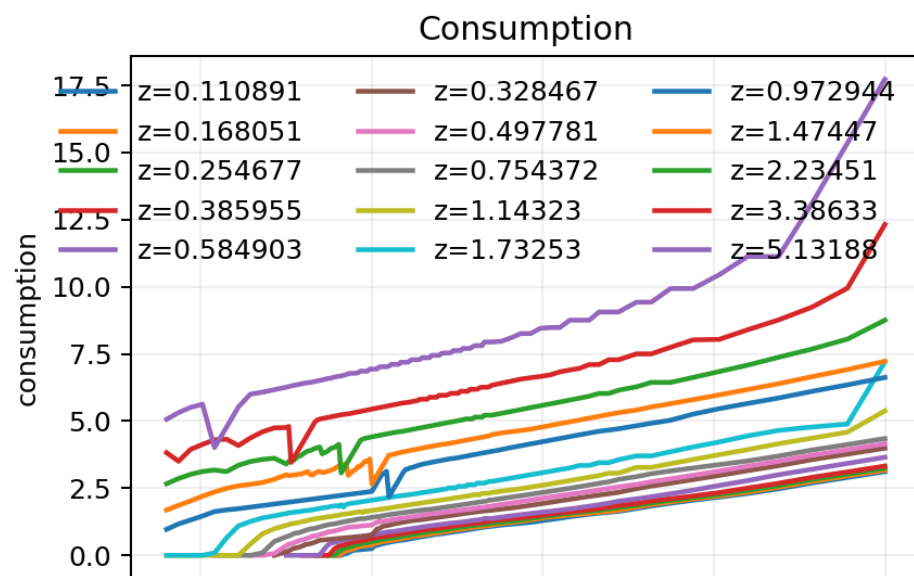
Fertility policy by age and income state



Fixed-parameter stationary policies: old utility

Pre-2007 stationary comparison; balanced pensions; original saved diagnostic, unchanged.

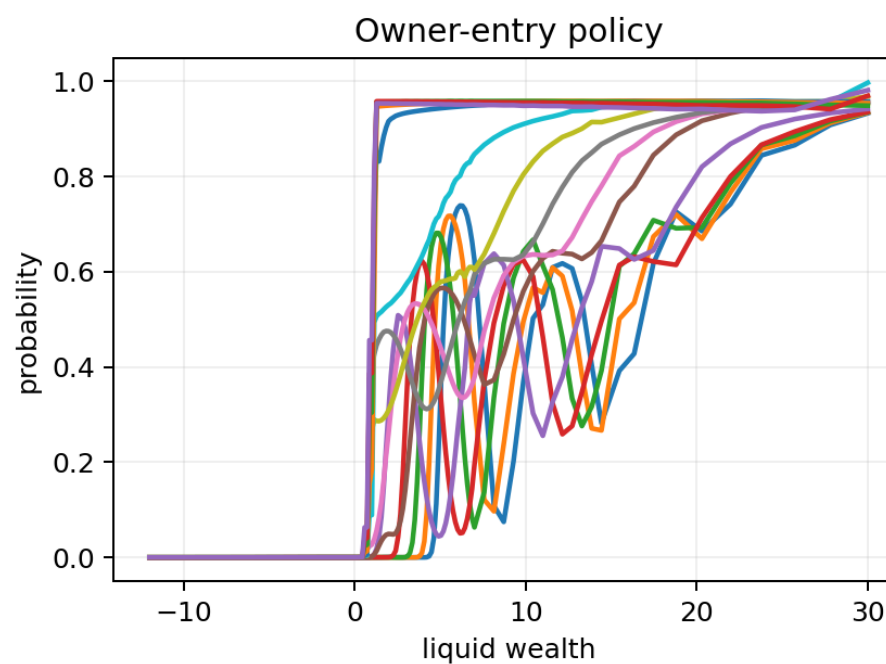
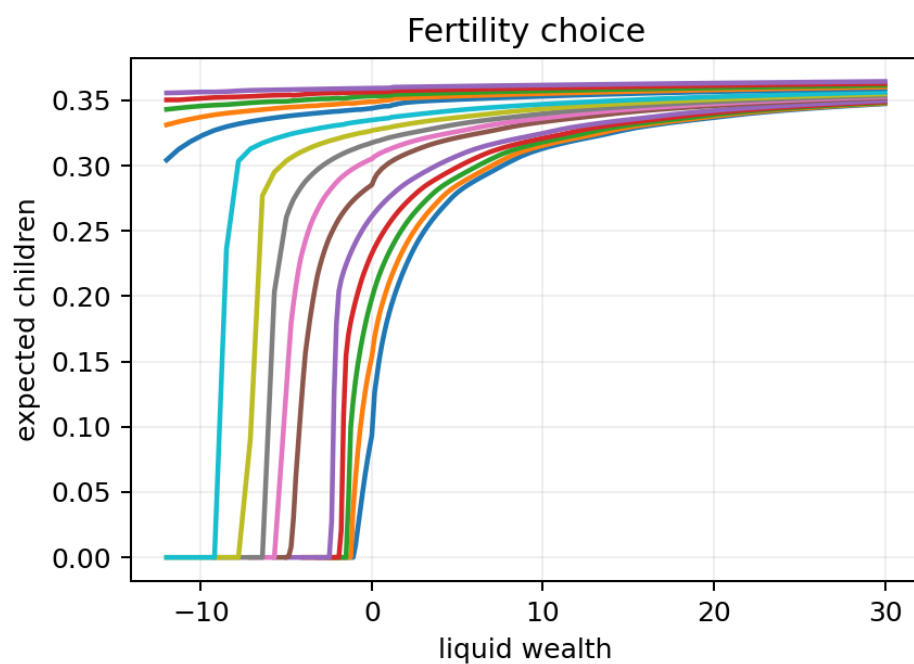
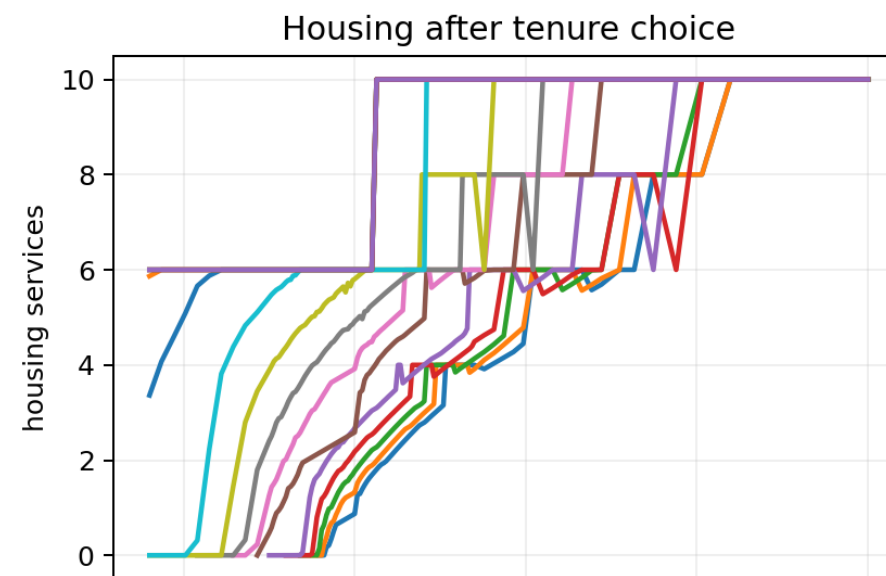
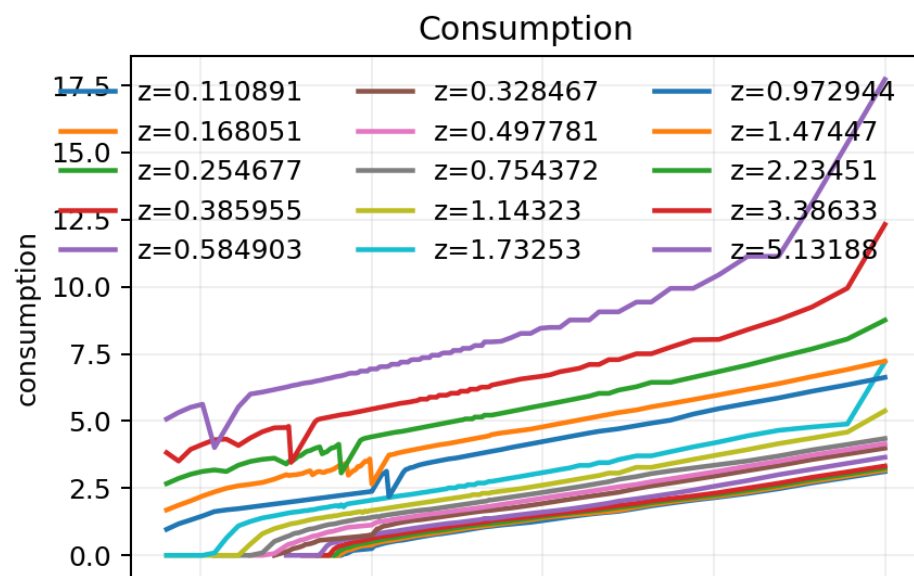
Childless renter policies at age 30



Fixed-parameter stationary policies: new utility

Pre-2007 stationary comparison; balanced pensions; original saved diagnostic, unchanged.

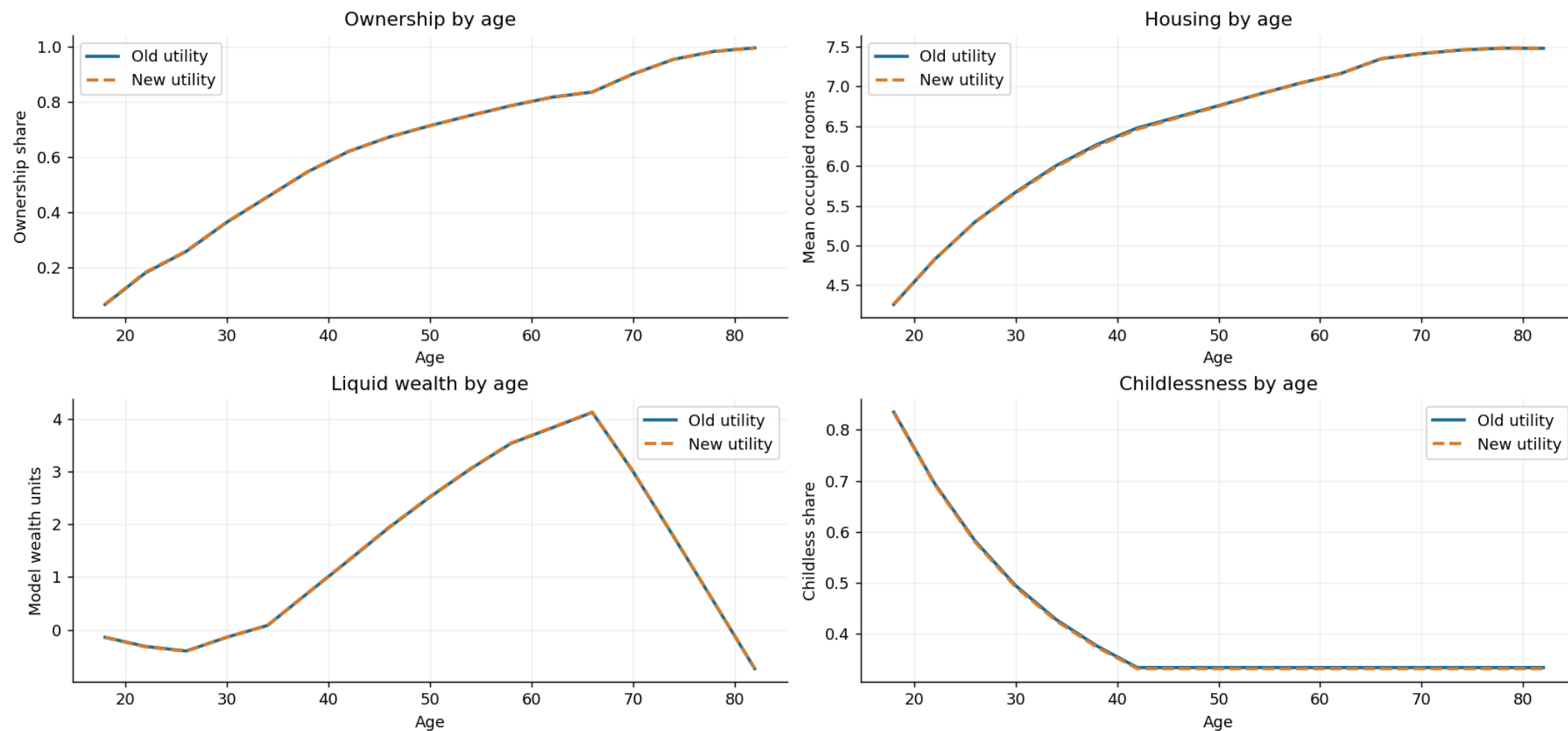
Childless renter policies at age 30



Stationary comparison at the retained 2023 preference

Both budgets balanced; unchanged structural parameters; no re-estimation or historical-population reproduction.

Old and new utility at the retained 2023 preference



Balanced pensions in both arms; same structural parameters and supply curve; two exact repetitions each.
Stationary counterparts, not recalibrated 2023 economies. They do not reproduce the historical 2023 population.